

**SUPPLEMENTARY MID TERM EXAMINATION**

B.TECH PROGRAMMES (UNDER THE AEGIS OF USICT) — THIRD SEMESTER, NOVEMBER 2024

**Paper Code: ES-201****Subject: Computational Methods****Time: 1½ Hrs.****Max. Marks: 30****Note: Attempt Q. No. 1, which is compulsory, and any two more questions from remaining.**

| Q. No. | Question                                                                                                                                                                                                                                                                                                                            | Marks | CO   |      |      |     |     |        |      |      |      |      |      |     |     |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|------|------|------|-----|-----|--------|------|------|------|------|------|-----|-----|
| 1(a)   | State Rolle's Theorem and check whether it is applicable for the function $f(x) =  x $ in $[-2, 2]$ or not?                                                                                                                                                                                                                         | [2.5] | CO1  |      |      |     |     |        |      |      |      |      |      |     |     |
| 1(b)   | How many steps of the bisection method are needed to find a root of the equation $x e^x = 1$ correct to three decimal places in the interval $[0, 1]$ ?                                                                                                                                                                             | [2.5] | CO1  |      |      |     |     |        |      |      |      |      |      |     |     |
| 1(c)   | Evaluate $\Delta^3[(1 - x)(1 - 2x)(1 - 3x)]$ where interval of differencing is 2.                                                                                                                                                                                                                                                   | [2.5] | CO2  |      |      |     |     |        |      |      |      |      |      |     |     |
| 1(d)   | Using Lagrange interpolation, find the unique polynomial of degree 2, such that $f(0) = 1$ , $f(1) = 3$ , $f(3) = 55$ .                                                                                                                                                                                                             | [2.5] | CO2  |      |      |     |     |        |      |      |      |      |      |     |     |
| 2(a)   | Find the real root of the equation $\cos x - x e^x = 0$ correct to four decimal places in $[0, 1]$ by using Newton-Raphson method.                                                                                                                                                                                                  | [5]   | CO1  |      |      |     |     |        |      |      |      |      |      |     |     |
| 2(b)   | For the following data obtain the backward differences interpolation polynomial and estimate $f(x)$ at $x = 0.35$ : <table><tr><td><math>x</math></td><td>0.1</td><td>0.2</td><td>0.3</td><td>0.4</td><td>0.5</td></tr><tr><td><math>f(x)</math></td><td>1.40</td><td>1.56</td><td>1.76</td><td>2.00</td><td>2.28</td></tr></table> | $x$   | 0.1  | 0.2  | 0.3  | 0.4 | 0.5 | $f(x)$ | 1.40 | 1.56 | 1.76 | 2.00 | 2.28 | [5] | CO2 |
| $x$    | 0.1                                                                                                                                                                                                                                                                                                                                 | 0.2   | 0.3  | 0.4  | 0.5  |     |     |        |      |      |      |      |      |     |     |
| $f(x)$ | 1.40                                                                                                                                                                                                                                                                                                                                | 1.56  | 1.76 | 2.00 | 2.28 |     |     |        |      |      |      |      |      |     |     |
| 3(a)   | Find the minimum of $f(x) = x(x - 1.5)$ in $[0, 1]$ using Fibonacci search algorithm by taking $n = 4$ .                                                                                                                                                                                                                            | [5]   | CO1  |      |      |     |     |        |      |      |      |      |      |     |     |
| 3(b)   | Evaluate $\int_0^3 (x^2 + 1) dx$ using Simpson's $\blacksquare$ rule and Simpson's $\blacksquare$ rule by taking $h = 1/2$ and compare the result with exact value.                                                                                                                                                                 | [5]   | CO2  |      |      |     |     |        |      |      |      |      |      |     |     |
| 4(a)   | By the method of steepest descent find the minimum value of the function $f(x_1, x_2) = x_1 - x_2 + 2x_1^2 + 2x_1x_2 + x_2^2$ starting from the point $X_0 = (0, 0)$ . Perform three iterations only.                                                                                                                               | [5]   | CO1  |      |      |     |     |        |      |      |      |      |      |     |     |
| 4(b)   | Given that $\log_{10} 654 = 2.8156$ , $\log_{10} 658 = 2.8182$ , $\log_{10} 659 = 2.8189$ , $\log_{10} 661 = 2.8202$ , find $\log_{10} 656$ using divided difference interpolation formula.                                                                                                                                         | [5]   | CO2  |      |      |     |     |        |      |      |      |      |      |     |     |

\*\*\*\* End of Question Paper \*\*\*\*